

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering

2. Subject Name: SIGNALS AND SYSTEMS L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: NIL

4. Syllabus and reference books:

Syllabus: The subject covers both the continuous-time and discrete-time signals and systems. Both time and frequency-domain analysis will be covered.

Reference Books:

1) Signals and Systems, Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab, Pearson Education India; 2nd edition, 2015.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Continuous-time (CT) and Discrete-time (DT) signals, Operation on signals, Sinusoid and Exponential Signals, Periodicity, Power and Energy, Unit Impulse, Step, and Ramp Functions, Measure of signals. Basic System Properties: Systems with and without Memory, Invertibility and Inverse Systems, Causality, Time Invariance and Linearity.	3
2.	DT and CT Linear Time-invariant (LTI) Systems, Representation of CT and DT Signals in terms of Impulse, Impulse Response of LTI systems and Convolution Integral, Properties of LTI Systems, Causal LTI Systems Described by Differential and Difference Equations and its Block Diagram and signal flow Representations	7
3.	Response of LTI Systems to Complex Exponentials, Fourier Series Representation of CT and DT Periodic Signals, Convergence and Properties of the Fourier Series, Fourier Series, Parseval's Relation, Induced gain of LTI systems, LTI systems with Periodic Inputs	6
4.	Representation of Aperiodic Signals: The CT and DT Fourier Transform, its Convergence and Examples, The Fourier Transform for Periodic Signals, Properties: Convolution, Multiplication and Duality	7
5.	A brief Overview of the Laplace Transform	1

6.	Transfer function, Concept of Poles and Zeros, Causality, Stability of LTI Systems, Block Diagram Representations of Causal LTI systems with Some Examples of Physical Systems	3
7.	The z-Transform, the Region of Convergence, Properties of the z-Transforms, Inverse z-transform, Initial- and Final-value Theorems in z-domain Transfer Function in z-Domain, Stability of LTI Systems in z-Domain, Block Diagram Representations of LTI Systems in z-Domain,	5
8.	Time Response of First-Order and Second-Order LTI Systems and Time-Domain Characterization- Rise time, Settling Time, Peak Overshoot, and Steady-state Error Specifications, Effect of Addition of Poles and Zeros - Concept of Dominant Poles and Zeros. Frequency Response of LTI Systems and Frequency-Domain Characterization, Asymptotic Bode (Magnitude and Phase) Plot	9
9.	Concept of Feedback Systems, Open-Loop and Closed-Loop Systems, Advantages of Feedback, Stability of a Feedback System	2
Total number of hours		43